GLOBAL ACADEMY OF TECHNOLOGY

Approved by AICTE, New Delhi, Recognized by the Govt. of Karnataka
Autonomous Institute affiliated to VTU, Belagavi,
NAAC Accredited with 'A' Grade
Ideal Homes Township, Rajarajeshwari Nagar, Bengaluru-98

Department of Computer Science and Engineering

A
Mini Project Report
on

"Superstore Sales Dashboard Using Power BI"

Submitted in partial fulfillment of Business Intelligence

with Mini project (22AML75A) of VIIth semester

Bachelor of Engineering

In

Computer Science and Engineering

Submitted by

Harshitha N K	**1GA22CS071**
Shreya N	**1GA22CS153**
Swara Shetty M	**1GA22CS173**
Swathi K N	**1GA22CS174**

Under the Guidance of

Prof. Sushmitha M
Assistant Professor
Dept. of AIML

GLOBAL ACADEMY OF TECHNOLOGY

Department of Artificial Intelligence and Machine Learning

Rajarajeshwari Nagar, Bengaluru – 560 098

CERTIFICATE

Certified that the VII Semester Mini Project in Business intelligence Mini Project Entitled **“Superstore Sales Dashboard Using Power BI”** carried out by **Ms. Harshitha N K** , bearing **USN 1GA22CS071, Ms. Shreya N**, bearing **USN 1GA22CS153, Ms. Swara Shetty M**, bearing **USN 1GA22CS173 and Ms. Swathi K N**, bearing **USN 1GA22CS174** is submitted in partial fulfillment for the award of the **Bachelor of Engineering** in Computer Science and Engineering during the year 2024-2025.The Business intelligence with Mini project report has been approved as it satisfies the academic requirements in respect of the mini project work prescribed for the said degree.

Prof. Sushmitha M
Assistant Professor,
Dept of AIML,
GAT, Bengaluru

GLOBAL ACADEMY OF TECHNOLOGY

Department of Artificial Intelligence and Machine Learning

Rajarajeshwari Nagar, Bengaluru – 560 098

DECLARATION

We, Harshitha N K, bearing USN 1GA22CS071, Shreya N, bearing USN 1GA22CS153, Swara Shetty M, bearing USN 1GA22CS173, Swathi K N, bearing USN 1GA22CS174 students of Seventh Semester B.E, Global Academy of Technology, Rajarajeshwari nagar Bengaluru, declare that the Mini Project entitled "***Superstore Sales Dashboard Using Power BI***" has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during the academic year 2025-2026.

Harshitha N K	**1GA22CS071**
Shreya N	**1GA22CS153**
Swara Shetty M	**1GA22CS173**
Swathi K N	**1GA22CS174**

Place: Bengaluru

Date:

ABSTRACT

The Superstore Sales Dashboard Project focuses on transforming raw sales data into meaningful business insights through data visualization and analysis. The dataset used contains detailed information on orders, products, customers, sales, profit, shipping methods, and payment modes across various regions and time periods. By building an interactive dashboard, this project enables a comprehensive understanding of the store's sales performance and profitability trends. Key performance indicators such as total sales, total profit, quantity sold, and average profit margin are highlighted for quick business evaluation. Various visual elements such as bar charts, pie charts, line graphs, and geographical maps illustrate critical metrics like sales by category, profit trends by month, preferred payment methods, and regional performance. The analysis reveals that the Office Supplies category contributes the highest sales, while Standard Class shipping and Cash on Delivery are the most preferred options. Seasonal spikes in sales during November and December suggest strong holiday demand patterns. Overall, the dashboard helps identify profitable segments, underperforming areas, and actionable strategies for improving business efficiency. This project demonstrates the importance of data analytics and visualization in supporting data-driven decision-making for retail operations.

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's data-driven business environment, organizations rely heavily on analytics to understand customer behavior and optimize sales strategies. Retail businesses, in particular, generate large volumes of transactional data that can provide valuable insights when properly analyzed. The Superstore Sales Dashboard Project aims to utilize this data to gain a clear understanding of the company's sales performance, profitability, and operational efficiency.

The dataset used in this project includes detailed attributes such as order date, shipping method, product category, region, payment mode, sales, quantity, and profit. Using visualization tools, these parameters are analyzed to discover trends, patterns, and correlations. The dashboard provides an interactive and user-friendly interface to explore sales and profit distribution across time, product categories, and regions.

The primary objective of the project is to convert complex data into easy-to-understand visuals that support strategic business decisions. Through effective data representation, the dashboard helps stakeholders monitor performance, identify best-selling products, analyze customer preferences, and evaluate the impact of seasonal changes. Ultimately, this project highlights how data visualization can bridge the gap between raw data and actionable business intelligence, making it a vital tool for modern retail management.

CHAPTER 2

REQUIREMENT SPECIFICATION

2.1 Software Requirements

1. Software Tools
    - Microsoft Power BI / Tableau / Excel Dashboard – For creating data visualizations and interactive dashboards.
    - Database / CSV File – The Superstore_Sales_Dataset.csv acts as the primary data source.
    - Operating System: Windows 10 or higher (or equivalent macOS/Linux environment).

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Objectives

The main objective of the Superstore Sales Dashboard Project is to analyze sales and profit data to gain valuable insights into business performance and support data-driven decision-making.

The specific objectives include:

1. To analyze overall sales performance – Understand total sales, profit, and quantity sold across different time periods.
2. To identify top-performing categories and sub-categories – Determine which product groups contribute most to sales and profitability.
3. To study customer preferences – Analyze purchasing trends based on segment, region, and payment mode.
4. To examine regional performance – Identify which states or regions generate higher sales and profit.
5. To evaluate shipping efficiency – Understand which shipping modes are most commonly used and profitable.
6. To observe time-based trends – Track monthly and yearly fluctuations in sales and profit to detect seasonal patterns.
7. To present data visually through a dashboard – Create an interactive and easy-to-understand dashboard using visualization tools.
8. To assist management in decision-making – Use insights to improve marketing strategies, inventory planning, and customer satisfaction.

CHAPTER 4

IMPLEMENTATION OF THE PROJECT

4.1 Implementation

The implementation of the Superstore Sales Dashboard Project involved several systematic steps from data preparation to visualization and analysis. The process was carried out using tools such as Microsoft Power BI and Microsoft Excel, with the dataset in CSV format serving as the input source. The steps followed are as below:

1. Data Collection
   The dataset named "SuperStore_Sales_Dataset.csv" was used. It contained detailed information about sales transactions including Order ID, Product Category, Sub-Category, Customer Segment, Region, Payment Mode, Sales, Quantity, Profit, and Shipping details.

2. Data Cleaning and Preprocessing
   Before visualization, the dataset was cleaned and formatted to ensure accuracy. This included:
   - Removing missing or duplicate entries.
   - Formatting date fields (Order Date, Ship Date).
   - Verifying data types for numerical and categorical values.
   - Ensuring consistency in region and category names.

3. Data Import
   The cleaned dataset was imported into Power BI for creating the dashboard. Relationships between different attributes were automatically detected or manually configured to support accurate visual analysis.

4. Data Visualization
   Multiple charts and graphs were created to highlight key business insights:
   - Cards: Displayed total sales, profit, and quantity for quick KPI overview.
   - Pie Charts: Represented distribution of sales by payment mode and customer segment.
   - Bar Charts: Showed category-wise and sub-category-wise sales performance.
   - Line Graphs: Illustrated monthly and yearly profit and sales trends.
   - Maps: Displayed regional distribution of sales and profit across U.S. states.

5. Dashboard Design
   The visuals were arranged in an interactive dashboard layout. Filters for Year and Region were added to allow users to analyze specific data segments. A consistent color scheme and clear labeling were used for better readability and professional appearance.

CHAPTER 5

RESULT

5.1 Snapshots

Fig 5.1: Sales Dashboard

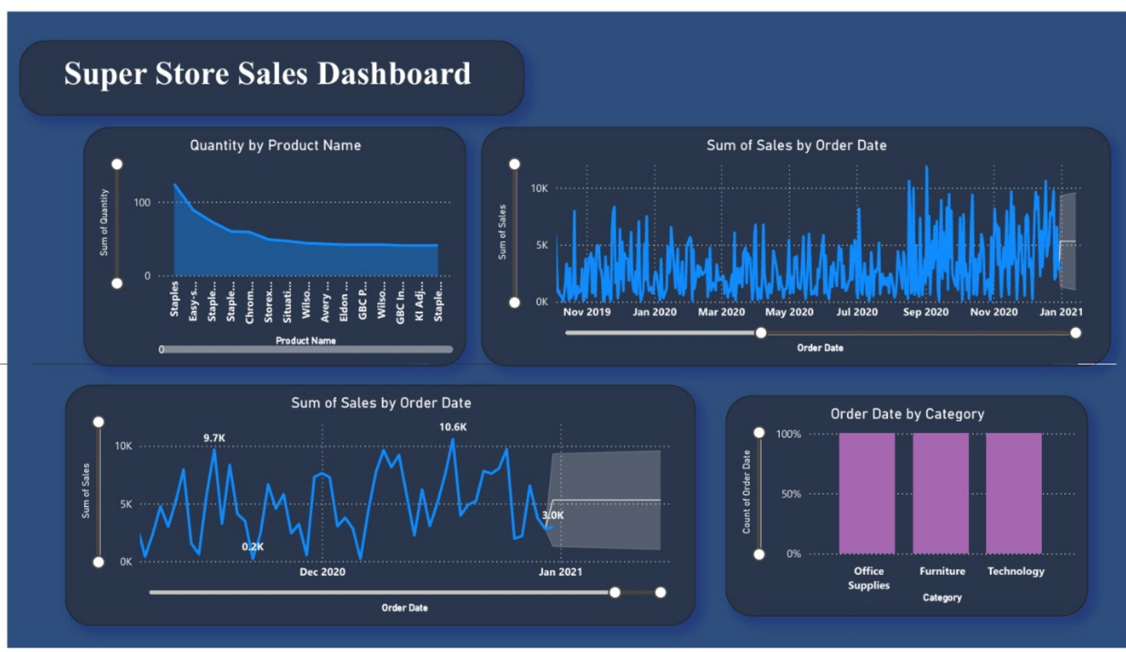

Fig 5.2: Forecasting of sales

CHAPTER 6

CONCLUSION

The Superstore Sales Dashboard Project successfully transformed raw sales data into meaningful business insights using Power BI. Through the integration of data visualization techniques, the dashboard provided a comprehensive overview of key performance indicators such as total sales, profit, and quantity sold, along with detailed analyses by region, category, sub-category, and payment mode. The project demonstrated how data analytics and visualization tools can simplify complex datasets and help businesses make data-driven decisions. Users can easily identify top-performing products, profitable regions, and customer preferences through interactive filters and charts. Overall, the dashboard serves as an efficient and user-friendly decision-support tool, enabling businesses to monitor performance, identify trends, and improve strategic planning. It highlights the power of Power BI in turning static data into actionable insights that enhance operational efficiency and profitability.